# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME: Artificial Intelligence** **COURSE CODE: CSA1741**

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**COURSE OUTCOME COVERED**

***CO1:*** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

*Assessment Tool Weightage: 10%*

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**QUESTIONS:2** **Total Marks:20**

## Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

***Code of Conduct:***

*I Niranjna devi S I / 192519100 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***
***Niranjna devi S I***

## RUBRICS FOR CALCULATION:

| Criterion | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) | Max Marks |
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| **Criteria** | **Level 4 (Exemplary)** | **Level 3 (Proficient)** | **Level 2 (Developing)** | **Level 1 (Beginning)** | **Max Marks** |
| Understanding of Concepts | Demonstrates thorough, accurate understanding of concepts | Good understanding with minor gaps | Basic understanding with some misconceptions | Poor understanding; major misconceptions | 5 |
| Explanation | Detailed, well-explained answers with relevant examples | Clear explanation with adequate details | Limited explanation; lacks depth | Poor or unclear explanation | 5 |
| Application | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors | Limited application with weak analysis | No application or incorrect analysis | 4 |
| Organization | Well-structured answers with logical flow and coherence | Organized with minor issues in flow | Basic structure, lacks clarity | Poor organization, incoherent answers | 3 |
| Accuracy | Completely accurate and covers all required points | Mostly accurate with minor omissions | Partially correct with missing points | Incorrect answers with major gaps | 2 |
| Clarity | Clear, neat, professional presentation | Understandable with minor issues | Limited clarity, readability issues | Poorly written, difficult to understand | 1 |

# Answer

## ***1.Smart Home Automation & Intelligent Agents:***

### 1.Detailed Breakdown of 4 Agent Types:

- **Simple Reflex Agent:**
  - **How it works:** Operates strictly on condition-action rules ( Eg: "If condition A, then do B"). It looks only at the current moment.
  - **Smart Home Example:** If a smoke detector trips, instantly sound the alarm. If motion is detected in the hallway, turn on the lights.

- **Model-Based Reflex Agent:**
  - **How it works:** Keeps an internal memory (state) of how the world works and how past actions affected it. It handles situations where sensors don't show the whole picture.
  - **Smart Home Example:** It tracks whether a window was left open 10 minutes ago, or remembers that the kitchen lights were manually turned off by the user so it shouldn't auto-turn them back on immediately.

- **Goal-Based Agent:**
  - **How it works:** Has explicit goals to achieve. It looks ahead and plans a sequence of actions to reach that target objective.
  - **Smart Home Example:** The goal is "House temperature must be 22°C by 7:00 AM." The agent calculates when to turn on the heater beforehand so it reaches the goal right on time.

- **Utility-Based Agent:**
  - **How it works:** Uses a "utility function" to measure how happy or satisfied the user is. When goals conflict, it chooses the option that gives the highest overall score.
  - **Smart Home Example:** Balancing maximum comfort (keeping the house warm) against minimum cost (saving electricity when rates are high during peak afternoon hours).

## 3. Functional Analysis Across Lighting, Temperature, and Security

- **Lighting Control:**
  - ***Simple Reflex:*** Turns lights on/off instantly via motion sensors.
  - ***Utility-Based:*** Adjusts brightness based on time of day, natural sunlight coming through windows, user presence, and electricity costs.

- **Temperature Control:**
  - ***Model-Based:*** Tracks how fast the house loses heat based on outdoor weather conditions.
  - ***Goal-Based/Utility-Based:*** Pre-cools or pre-heats the house before the user arrives home, while choosing the cheapest energy source.

- **Security Control:**
  - ***Simple Reflex:*** Triggers immediate locks or alarms when a door sensor breaks (safety-critical, zero delay required).
  - ***Model-Based:*** Differentiates between a family member returning late and an actual intruder based on historical entry patterns.

## 4. Detailed Justification for the Best Approach

- **The Winning Choice:** A Hybrid Architecture (Utility-Based Core with Simple Reflex Safety Overrides).

- **Why this combination works best:**
  - **For Safety & Security:** You cannot wait for calculations or planning when a fire starts or a burglar breaks in. Simple reflex rules give instant, life-saving reactions.
  - **For Comfort & Energy:** Running a smart home efficiently requires continuous trade-offs between cost and comfort. A utility-based model learns user preferences over weeks, predicts when rooms will be used, and cuts down power bills without sacrificing comfort.

## ***2.Robot in an Unknown Maze & Search Strategies:***

### 1. How Uninformed Search Works in an Unknown Maze

Since the robot starts with zero knowledge of the maze walls, dead ends, or exit location, it must explore blindly:

- **Uniform Cost Search (UCS):**
  - Explores paths by expanding the lowest cumulative cost node first.
  - In a maze where every step costs 1, it expands outward evenly in all directions like expanding ripples in water.

- **Iterative Deepening Search (IDS):**
  - Runs a series of Depth-First Searches with increasing depth limits (Limit = 1, then 2, then 3, etc.).
  - It explores deep into a hallway, hits a dead end, backs up, and tries the next path safely within memory limits.

### 2. In-Depth Pros, Cons, and Complexities

- **Uniform Cost Search (UCS):**
  - *Pros:* Guaranteed to find the absolute shortest path to the exit. Works with varying step costs (e.g., mud vs. smooth floor).
  - *Cons:* Consumes massive amounts of memory because it stores every single visited path in a queue. In a branching maze, it quickly causes memory overflow on small robot hardware.

- **Iterative Deepening Search (IDS):**
  - *Pros:* Very memory-efficient ($O(bd)$). Perfect for robots with limited RAM and processing power. Complete and optimal for unit step costs.
  - *Cons:* Repeats searching the upper levels of the tree multiple times, which wastes some CPU time, but saves critical memory.

### 3. Agent Design & Decision-Making Dynamics

- **Agent Classification:** The robot functions as a **Goal-Based Problem-Solving Agent**. Its sole objective is reaching the exit state.

- **Online vs. Offline Search:**
  - Because the maze is *unknown*, **offline search** (planning the whole route beforehand) is impossible.
  - The robot must use **online search**: it takes a step, uses its sensors to see nearby walls, updates its internal map, and makes its next decision step-by-step in real time.

**4. Justification of the Best Combination**

- **The Winning Choice:** A **Goal-Based Agent paired with Iterative Deepening Search (IDS)**.

- **Why this choice wins:**
  - Robots have tight physical limits on battery and onboard computer memory.
  - While UCS finds the shortest path, its high memory usage risks crashing the robot's system mid-exploration.
  - **IDS** provides the safety of finding the exit while keeping memory usage extremely low, making it the most practical and reliable choice for real-world hardware navigating unknown spaces.